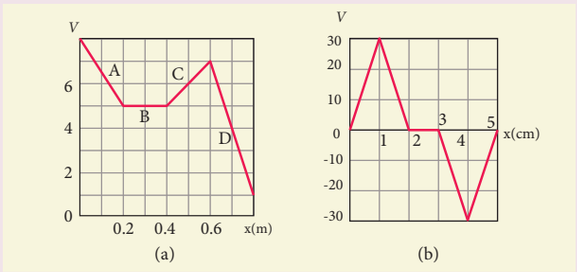


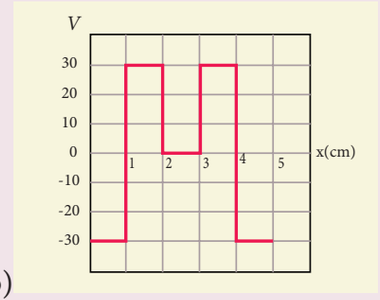
Calculate the electric flux through the (a) vertical rectangular surface (b) slanted surface and (c) entire surface.

Ans: (a) 15 $Nm^2 C^{-1}$ (b) 15 $Nm^2 C^{-1}$ (c) zero

8. The electrostatic potential is given as a function of x in figure (a) and (b). Calculate the corresponding electric fields in regions A, B, C and D. Plot the electric field as a function of x for the figure (b).



Ans: (a) E_X = 15 Vm^{-1} (region A), E_X = -10 Vm^{-1} (region C)
E_X = 0 (region B), E_X = 30 Vm^{-1} (region D)



(b)

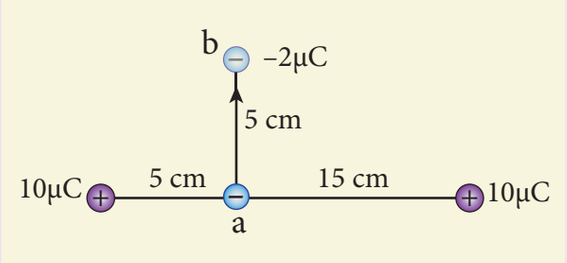
9. A spark plug in a bike or a car is used to ignite the air-fuel mixture in the engine. It consists of two electrodes separated by a gap of around 0.6 mm gap as shown in the figure.



To create the spark, an electric field of magnitude 3 × 10^6 Vm^{-1} is required. (a) What potential difference must be applied to produce the spark? (b) If the gap is increased, does the potential difference increase, decrease or remains the same? (c) find the potential difference if the gap is 1 mm.

Ans: (a) 1800 V, (b) increases (c) 3000 V

10. A point charge of +10 μC is placed at a distance of 20 cm from another identical point charge of +10 μC. A point charge of -2 μC is moved from point a to b as shown in the figure. Calculate the change in potential energy of the system? Interpret your result.



Ans: ΔU = -3.246 J, negative sign implies that to move the charge -2μC no external work is required. System spends its stored

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